

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

May 2012

EC 311 VLSI Technology & Design

Date: 10.05.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed

SECTION - I

- Q - 1 (a) Explain flat band voltage for p-type doped silicon substrate MOS structure. [02]
(b) Explain the fabrication process of nMOS transistor in detail. [05]
- Q - 2 (a) What is Gradual Channel Approximation (GCA)? With help of that derive current voltage equations for nMOS transistor. Further plot I-V characteristic for the same. [07]
(b) Discuss VLSI design flow in brief. [05]
(c) Why device isolation is required? [02]

OR

- Q - 2 (a) Explain nMOS system under external bias in detail with necessary energy band diagrams. [07]
(b) Derive the threshold voltage of two input CMOS- NOR gate. [05]
(c) Compare inverter circuit with saturated enhancement type nMOS load and inverter circuit with linear enhancement type nMOS load in brief. [02]
- Q - 3 (a) Draw and explain the inverter circuit with resistive load. Derive critical voltage points V_{OL} , V_{IL} , V_{OH} , V_{IH} . [07]
(b) Consider the following p-channel MOSFET process: [05]

Substrate doping $N_D = 10^{15} \text{ cm}^{-3}$ Polysilicon gate doping density $N_D = 10^{20} \text{ cm}^{-3}$, gate oxide thickness $t_{ox} = 650 \text{ \AA}$, and oxide-interface charge density $N_{ox} = 2 \times 10^{10} \text{ cm}^{-2}$. Use $\epsilon_{ox} = 3.97 \epsilon_0$, $\epsilon_{si} = 11.7 \epsilon_0$ where, $\epsilon_0 = 8.85 \times 10^{-14} \text{ F/cm}$

- (a) Calculate the threshold voltage V_{TO} , for $V_{SB} = 0$.
(b) Determine the type and the amount of channel ion implantation which are necessary to achieve a threshold voltage of $V_{TO} = -2 \text{ V}$.
(c) Explain subthreshold current and punch through effect in small geometry MOSFET. [02]

OR

- Q - 3 (a) What is the need of Voltage Boot strapping Circuit? Draw Dynamic bootstrapping arrangement and explain it in detail. [07]
- (b) Explain Dynamic CMOS Logic (Precharge-Evaluate Logic) in detail. Also illustrate the cascading problem in dynamic CMOS logic. [07]

SECTION – II

- Q - 4 (a) Prove that switching power dissipation of CMOS inverter circuit is proportional to switching frequency. [04]
- (b) Explain designing issues of CMOS inverter in detail. [03]
- Q - 5 (a) Explain Latch-up problem in CMOS inverter. Mention causes of latch-up and guidelines for avoiding latch-up. [07]
- (b) Explain the basic principles of pass transistor circuits. Also explain logic '1' and logic '0' transfer. [07]

OR

- Q - 5 (a) Explain CMOS D latch and Edge-Triggered flip flop with its timing diagrams. [07]
- (b) Explain Ad-hoc testable design techniques in detail. [07]
- Q - 6 (a) Explain Built-in Self test (BIST) techniques in brief. [05]
- (b) Explain Generalized NAND structure with Multiple inputs using combinational MOS logic circuits. [04]
- (b) Explain CMOS ring oscillator circuit in detail. [05]

OR

- Q - 6 (a) Draw input and output waveforms during high to low transition of output for a CMOS inverter and derive the expression for τ_{PHL} . [07]
- (b) Compare the two technology scaling methods, namely, (i) the constant electric-field scaling and (ii) the constant power-supply voltage scaling. In particular, show analytically by using equations how the delay time, power dissipation and power density are affected in terms of the scaling factor, S. [07]

To be more specific, what would happen if the design rules change from, say, $1\ \mu\text{m}$ to $1/S\ \mu\text{m}$ ($S > 1$)?
